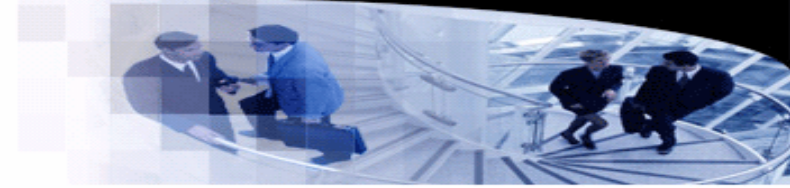




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Programming Languages and Compilers

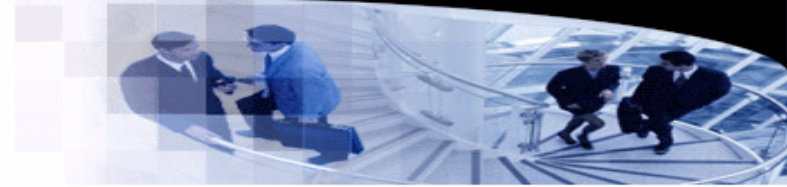
Lecture 23. Reaching Definitions

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Data Flow Equations



- **Control flow graph (CFG)**

- A graph form representation of a program.
- A vertex represents a basic block.
- An edge (s, t) indicates that t may be executed right after s .

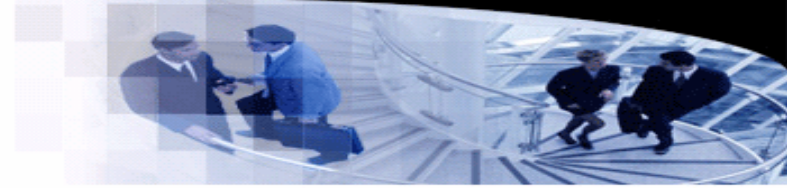
- **Data flow analysis**

- An approach to calculate some properties via examining how data flows CFGs.
- As CFGs are complicated, it is hard to claim these properties directly.

- **Data flow equation**

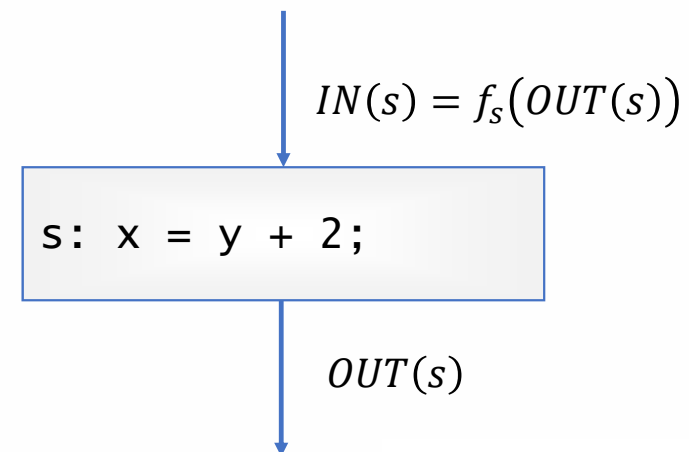
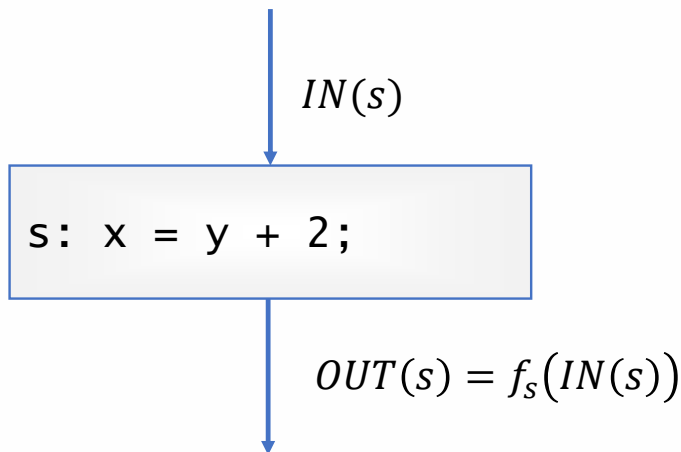
- For each statement or basic block in the CFG, we write an equation to describe how this statement processes the data.
 - The equation may describe the constraints over the input data and the output data.
 - We put these equations together and try to find a solution for them.
 - The solution leads to the claim of properties.
-

Data Flow Equations

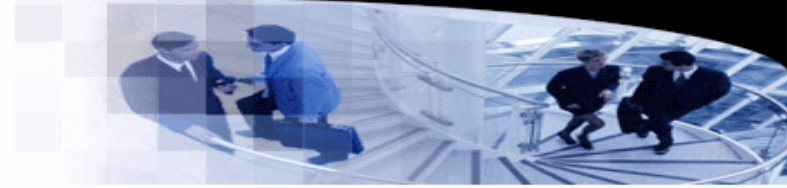


- Statements as processing functions

- Given a statement s , $IN(s)$ and $OUT(s)$ denote the data-flow values before and after the processing of s .
- Data flow equations on statements
 - Forward: $OUT(s) = f_s(IN(s))$
 - Backward: $IN(s) = f_s(OUT(s))$



Reaching Definition Equations



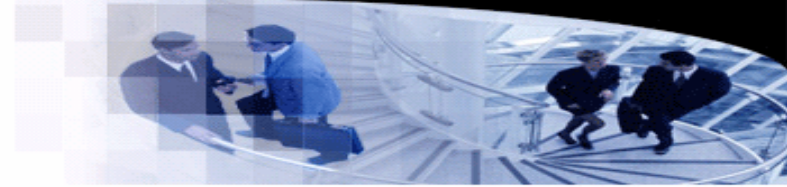
- **Facts of a statement**

- Consider a statement S : “ $a = b + c$ ”.
- It **uses** two variables, i.e., b and c .
- It **kills** all old definitions of a .
- It **generates** a new definition of a .

- **Data flow values (for reaching definition analysis)**

- Therefore, the data value can be represented in a set of definitions: $D = \{(v, p) | v \text{ is defined at statement } p.\}$.
- Since a statement only defines one variable in this form, the data flow value can be simplified as $D = \{p | p \text{ defines a variable that reaches this statement.}\}$.

Reaching Definition Equations



- Gen and kill on statements

- Given an input data flow value, which is a set of statement locations that defines variables, the process of a statement can be transformed into to sets, i.e., gen and kill.
- The gen set indicates that the statement generates a set of definitions.
 - $gen_s = \{s\}$
- The kill set indicates that the statement kills a set of definitions.
 - If the statement defines a variable v , the kill set includes all possible statement locations that defines v .
 - $kill_s = \{p \mid p \neq s \text{ and the statement } p \text{ defines } a.\}$

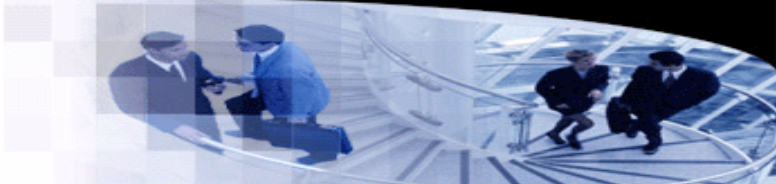
- Data flow equations over statements

- Let $IN(s) = \{p \mid p \text{ defines a variable that reaches } s.\}$
- We have $OUT(s) = (IN(s) - kill_s) \cup gen_s$.

Reaching Definition Equations

- Gen and kill on statements

- Consider the program on the right hand side.
- Statements with no definitions are ignored.
- There are 7 statements, each of which defines a variable.
- Consider the statement at d7.
- $gen_{d7} = \{d7\}$, as d7 defines one variable i .



```
ENTRY:
d1: i = m - 1
d2: j = n
d3: a = u1
    br d4
d4: i = i + 1
d5: j = j - 1
    br d6, d7
d6: a = u2
    br d7
d7: i = u3
    br d4, EXIT
EXIT:
```

Reaching Definition Equations

- Gen and kill on statements

- Consider the program on the right hand side.
- Statements with no definitions are ignored.
- There are 7 statements, each of which defines a variable.
- Consider the statement at d7.
- $gen_{d7} = \{d7\}$, as d7 defines one variable i .
- Two other statements (i.e., d1 and d4) also define i .

ENTRY:

d1: $i = m - 1$

d2: $j = n$

d3: $a = u1$
br d4

d4: $i = i + 1$

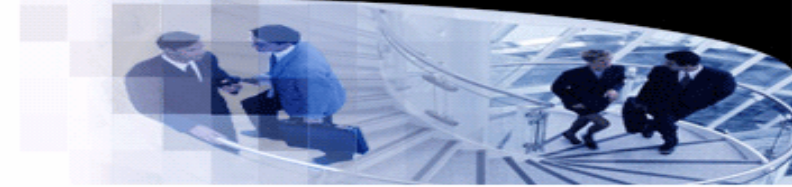
d5: $j = j - 1$
br d6, d7

d6: $a = u2$
br d7

d7: $i = u3$

br d4, EXIT

EXIT:



Reaching Definition Equations

- Gen and kill on statements

- Consider the program on the right hand side.
- Statements with no definitions are ignored.
- There are 7 statements, each of which defines a variable.
- Consider the statement at d7.
- $gen_{d7} = \{d7\}$, as d7 defines one variable i .
- Two other statements (i.e., d1 and d4) also define i .
- Therefore, $kill_{d7} = \{d1, d4\}$.

ENTRY:

d1: $i = m - 1$

d2: $j = n$

d3: $a = u1$
br d4

d4: $i = i + 1$

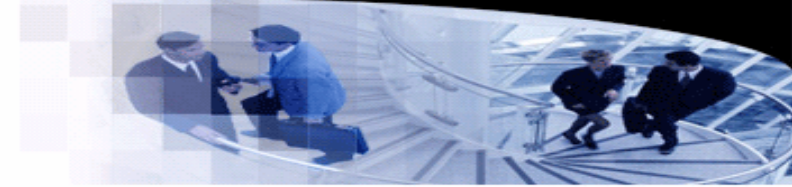
d5: $j = j - 1$
br d6, d7

d6: $a = u2$
br d7

d7: $i = u3$

br d4, EXIT

EXIT:



Reaching Definition Equations

- Gen and kill on statements

- Consider the program on the right hand side.
- Statements with no definitions are ignored.
- There are 7 statements, each of which defines a variable.
- Consider the statement at d7.
- $gen_{d7} = \{d7\}$, as d7 defines one variable i .
- Two other statements (i.e., d1 and d4) also define i .
- Therefore, $kill_{d7} = \{d1, d4\}$.
- We have the equation $OUT(d7) = (IN(d7) - \{d1, d4\}) \cup \{d7\}$.

ENTRY:

d1: $i = m - 1$

d2: $j = n$

d3: $a = u1$
br d4

d4: $i = i + 1$

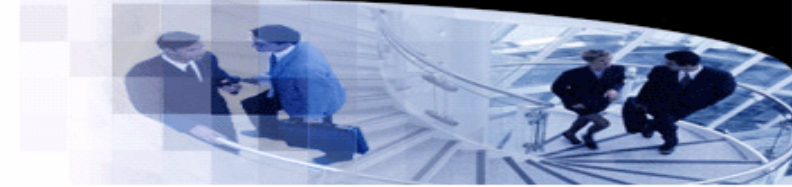
d5: $j = j - 1$
br d6, d7

d6: $a = u2$
br d7

d7: $i = u3$

br d4, EXIT

EXIT:



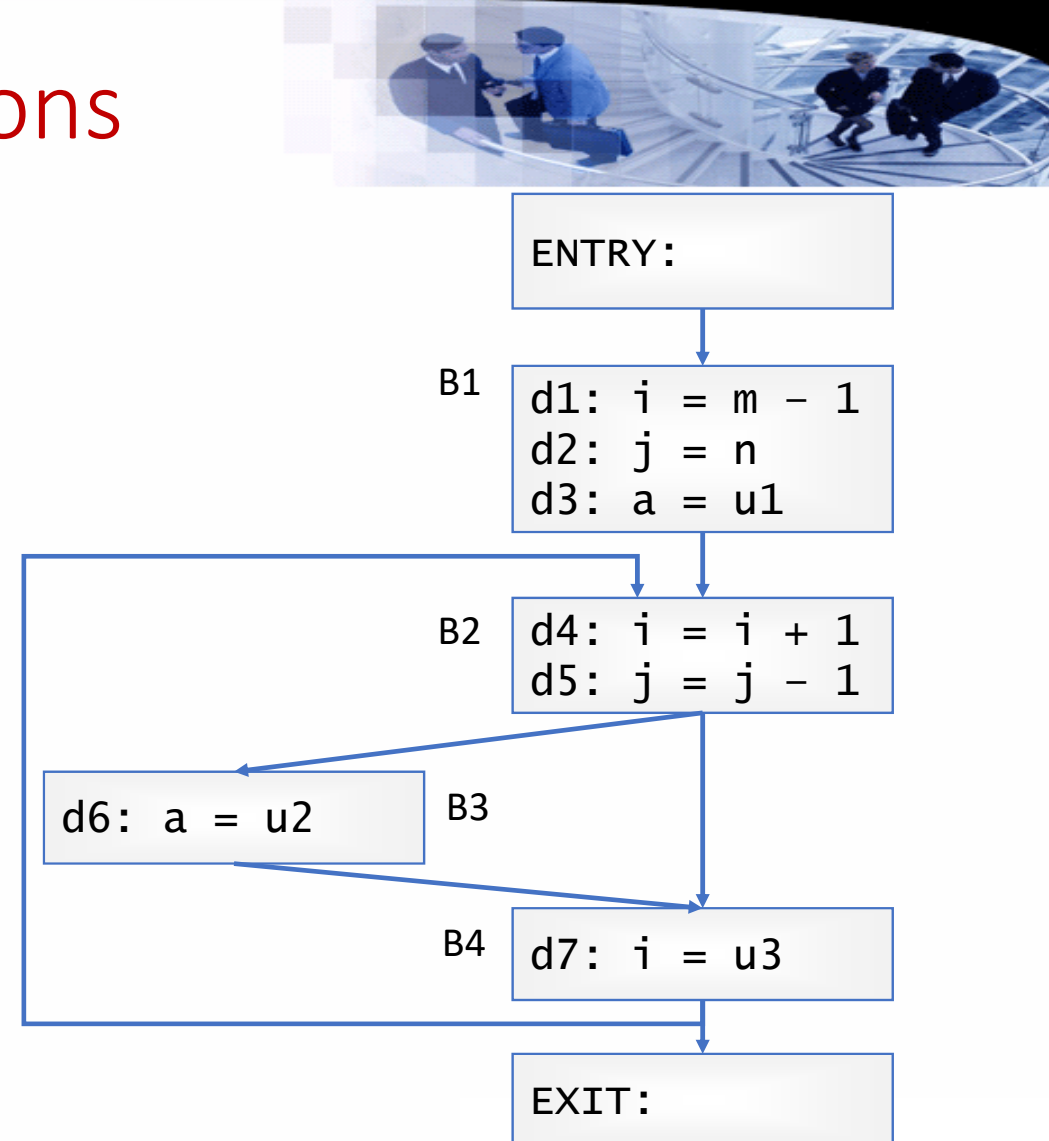
Reaching Definition Equations

- Gen and kill on basic blocks

- As the program is represented in a CFG, the equation is presented on basic blocks.
- Let a basic block $B = s_1; s_2; \dots; s_n$.
- $gen_B = \bigcup_{i=1}^n gen_{s_i}$.
- $kill_B = \bigcup_{i=1}^n kill_{s_i}$.

- Example

- Consider the basic block B2.
- $gen_{B2} = \{d4, d5\}$.
- $kill_{B2} = \{d1, d2, d7\}$.



Reaching Definition Equations

- Equations over basic blocks

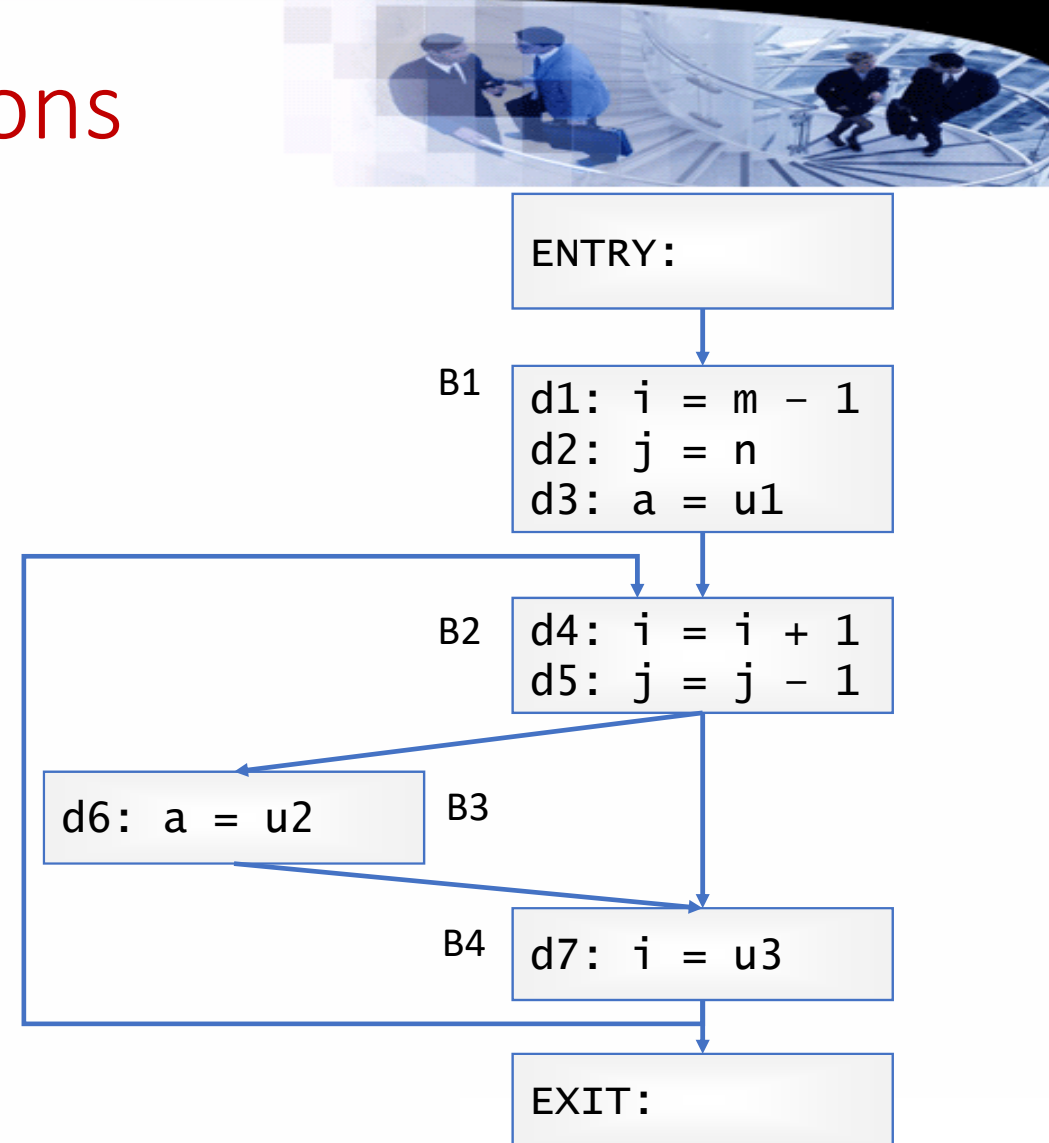
- $OUT(B) = (IN(B) - kill_B) \cup gen_B.$

- Equations over control flows

- $IN(B) = \bigcup_{P \in prev(B)} OUT(P)$

- Example

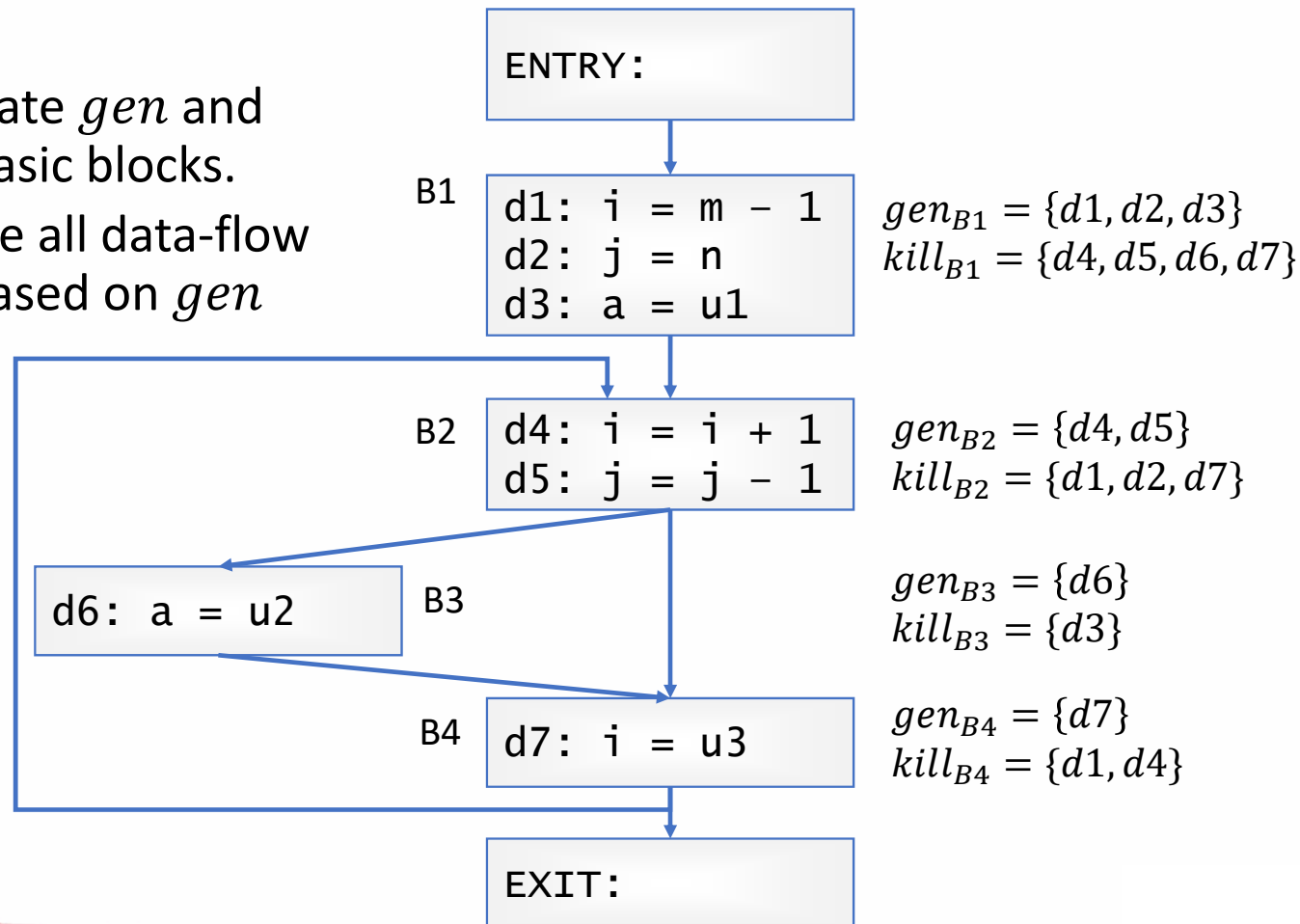
- Consider the basic block B2.
- $OUT(B2) = (IN(B2) - \{d1, d2, d7\}) \cup \{d4, d5\}.$
- $IN(B2) = OUT(B1) \cup OUT(B4).$



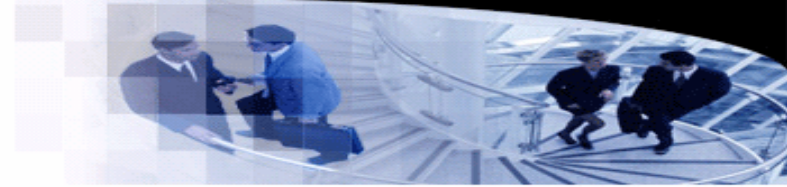
Reaching Definition Equations

• Example

- Let us calculate *gen* and *kill* for all basic blocks.
- Now we have all data-flow equations based on *gen* and *kill*.



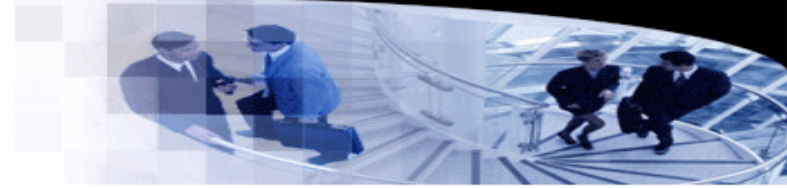
Solving Equations



- **Fixed-point problem**

- Given an arbitrary control-flow graph, the *IN* and *OUT* sets in the equations may depend on each other.
- It is challenging to solve the equations.
- If we can transform the equations into a fixed-point problem, it is possible to solve the problem via iterations.
- **Observation:**
 - As $IN(B) = \bigcup_{P \in prev(B)} OUT(P)$, the *IN* sets can be replaced by *OUT* sets.
- Given the set of all basic blocks $B_{all} = \{B_1, B_2, \dots, B_n\}$, we define a point $x = (OUT(B_1), OUT(B_2), \dots, OUT(B_n))$.
- The solution of the equations is the **least** fixed point of $x = f(x)$, where f generates a new set of OUTs by performing a forward reaching definition calculation.
 - For each B^i in x^i , we calculate $OUT^{i+1}(B)$ by $OUT^{i+1}(B) = (IN^{i+1}(B) - kill_B) \cup gen_B$.

Solving Equations



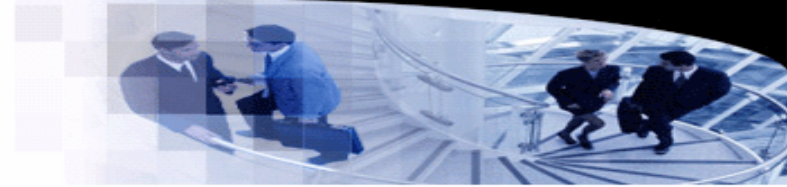
- **Fixed-point problem**

- **Knaster-Tarski theorem:** Let S be a set $\{s_0, s_1, \dots, s_n\}$ with $n + 1$ elements. If $f: \mathcal{P}(S) \rightarrow \mathcal{P}(S)$ is a monotone function, then $f^{n+1}(\emptyset)$ is the least fixed point of f and $f^{n+1}(S)$ is the greatest fixed point of f .
- Implications for calculating reaching definitions.
 - We can start the computation from $\emptyset$, which means $\forall i (OUT(B_i) = \emptyset)$.
 - The iteration stops at most $k \times n$ steps, where k is number of basic blocks and n stands for the number of definition statements.
- Hence, the reaching definition problem is tractable.

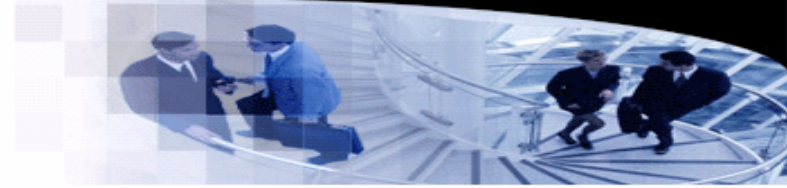
Solving Equations

- Algorithm

- $OUT(entry) = \emptyset$
- **foreach** basic block B **do**
- $OUT(B) = \emptyset$
- **while** OUT changed **do**
- **foreach** basic block B **do**
- $IN(B) = \bigcup_{P \in prev(B)} OUT(P)$
- $OUT(B) = gen_B \cup (IN(B) - kill_B)$
- **endfor**
- **endwhile**



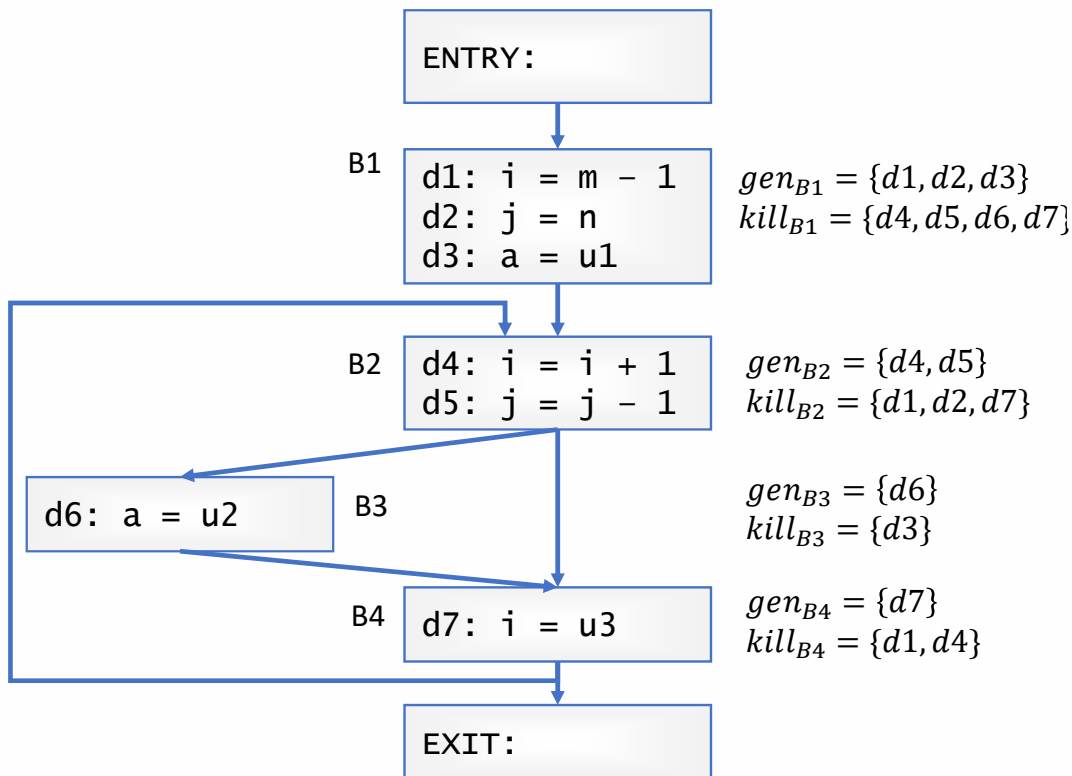
Solving Equations



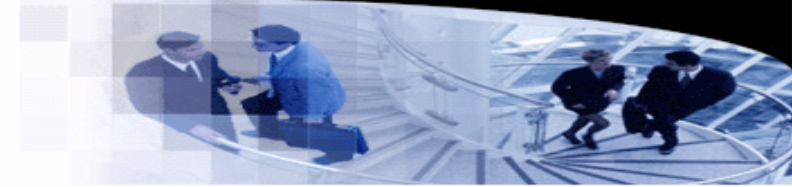
- **Bit vector representations for reaching definitions**
 - Given a data-flow value of reaching definition $OUT(B) = \{p_1, p_2, \dots, p_k\}$, the bit vector representation $V_{OUT}(B)$ is defined as follows.
 - If a statement $s_i \in OUT(B)$, where i is the index of the statement, then $V_{OUT}(B)[i] = 1$.
 - If $s_i \notin OUT(B)$, then $V_{OUT}(B)[i] = 0$.
- **Example**
 - Let us consider a bit vector $V_{OUT}(B) = (1, 0, 0, 0, 1, 0, 0)$.
 - It means that the definition at s_1 and s_5 reach the exit point of the basic block B .

Solving Equations

- Example

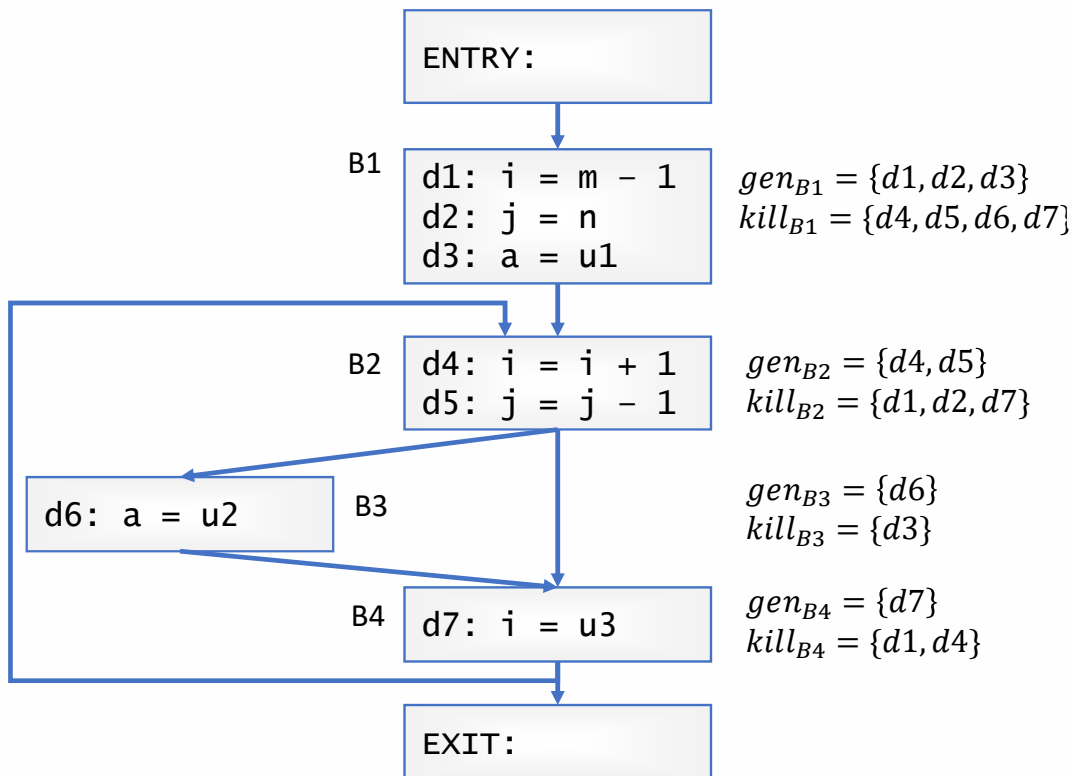


Block B	OUT[B] ⁰	IN[B] ¹	OUT[B] ¹	IN[B] ²	OUT[B] ²
B1	000 0000	000 0000	111 0000	000 0000	111 0000
B2	000 0000	111 0000	001 1100	111 0111	001 1110
B3	000 0000	001 1100	000 1110	001 1110	000 1110
B4	000 0000	001 1110	001 0111	001 1110	001 0111
EXIT	000 0000	001 0111	001 0111	001 0111	001 0111



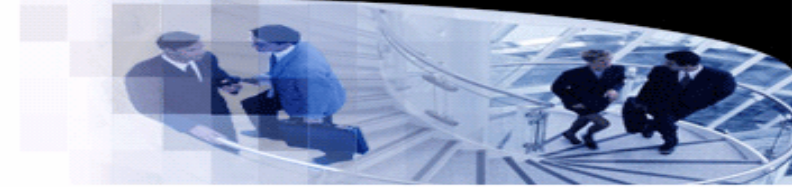
Solving Equations

- Example



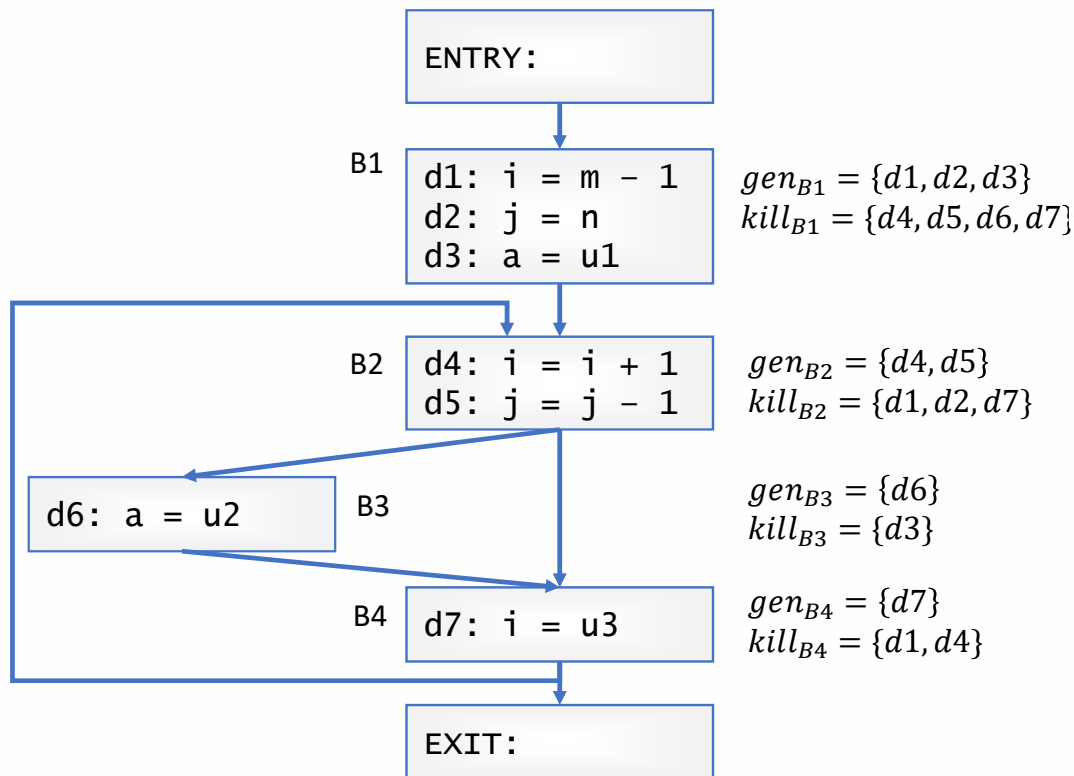
Block B	OUT[B] ⁰	IN[B] ¹	OUT[B] ¹	IN[B] ²	OUT[B] ²
B1	000 0000	000 0000	111 0000	000 0000	111 0000
B2	000 0000	111 0000	001 1100	111 0111	001 1110
B3	000 0000	001 1100	000 1110	001 1110	000 1110
B4	000 0000	001 1110	001 0111	001 1110	001 0111
EXIT	000 0000	001 0111	001 0111	001 0111	001 0111

$$IN(B1) = OUT(ENTRY) = 000\ 0000$$



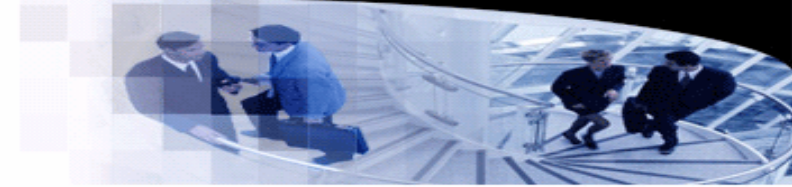
Solving Equations

- Example



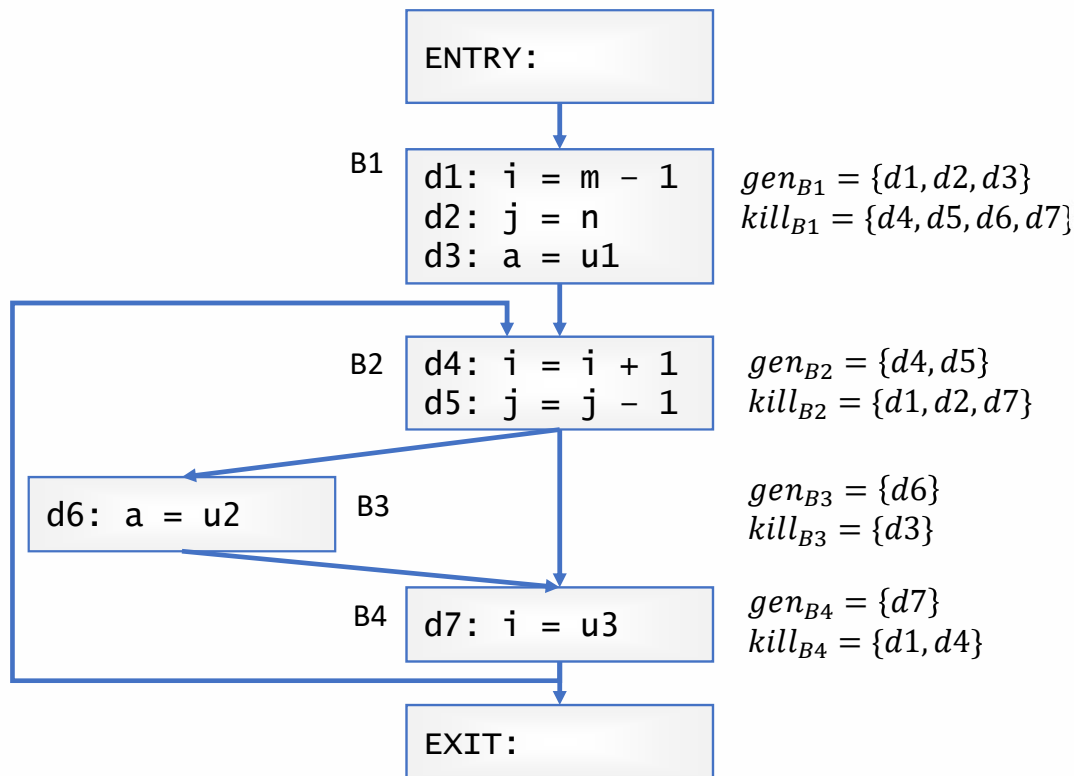
Block B	OUT[B] ⁰	IN[B] ¹	OUT[B] ¹	IN[B] ²	OUT[B] ²
B1	000 0000	000 0000	111 0000	000 0000	111 0000
B2	000 0000	111 0000	001 1100	111 0111	001 1110
B3	000 0000	001 1100	000 1110	001 1110	000 1110
B4	000 0000	001 1110	001 0111	001 1110	001 0111
EXIT	000 0000	001 0111	001 0111	001 0111	001 0111

$$OUT(B1) = IN(B1) - \{d4, d5, d6, d7\} \cup \{d1, d2, d3\} = 111\ 0000$$



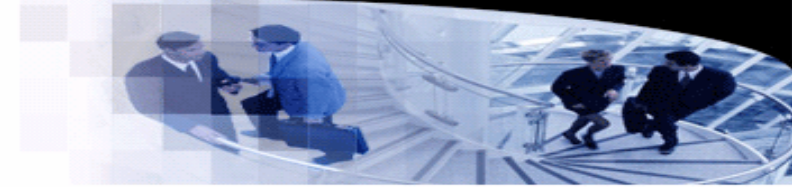
Solving Equations

- Example



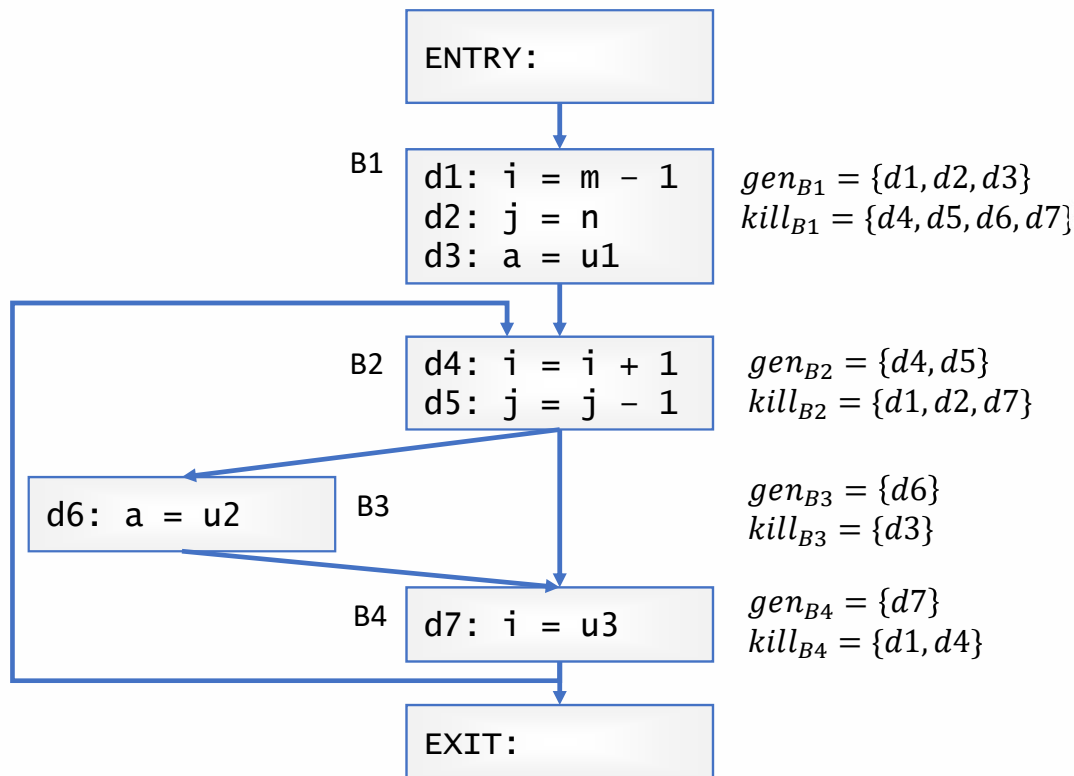
Block B	OUT[B] ⁰	IN[B] ¹	OUT[B] ¹	IN[B] ²	OUT[B] ²
B1	000 0000	000 0000	111 0000	000 0000	111 0000
B2	000 0000	111 0000	001 1100	111 0111	001 1110
B3	000 0000	001 1100	000 1110	001 1110	000 1110
B4	000 0000	001 1110	001 0111	001 1110	001 0111
EXIT	000 0000	001 0111	001 0111	001 0111	001 0111

$$IN(B2) = OUT(B1) \cup OUT(B4) = 111\ 0000$$



Solving Equations

- Example

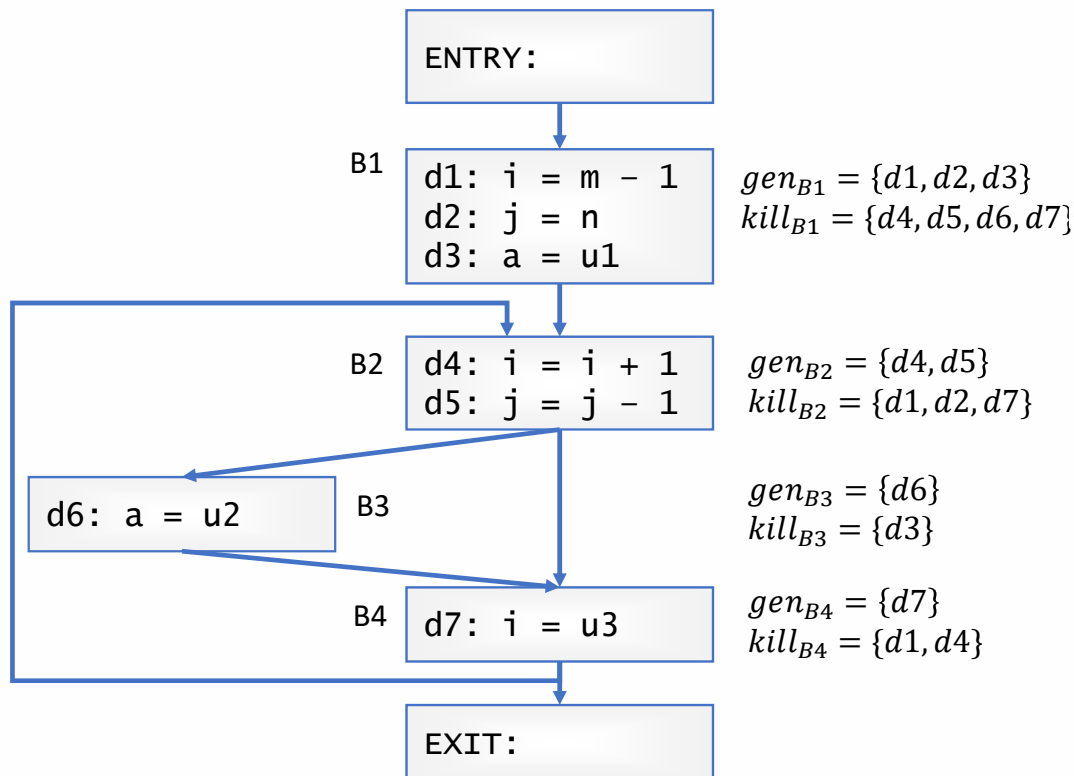


Block B	OUT[B] ⁰	IN[B] ¹	OUT[B] ¹	IN[B] ²	OUT[B] ²
B1	000 0000	000 0000	111 0000	000 0000	111 0000
B2	000 0000	111 0000	001 1100	111 0111	001 1110
B3	000 0000	001 1100	000 1110	001 1110	000 1110
B4	000 0000	001 1110	001 0111	001 1110	001 0111
EXIT	000 0000	001 0111	001 0111	001 0111	001 0111

$$\begin{aligned}
 OUT(B2) &= IN(B2) - \{d1, d2, d7\} \cup \{d4, d5\} \\
 &= 001\ 0000 \cup \{d4, d5\} \\
 &= 001\ 1100
 \end{aligned}$$

Solving Equations

- Example



Block B	OUT[B] ⁰	IN[B] ¹	OUT[B] ¹	IN[B] ²	OUT[B] ²
B1	000 0000	000 0000	111 0000	000 0000	111 0000
B2	000 0000	111 0000	001 1100	111 0111	001 1110
B3	000 0000	001 1100	000 1110	001 1110	000 1110
B4	000 0000	001 1110	001 0111	001 1110	001 0111
EXIT	000 0000	001 0111	001 0111	001 0111	001 0111

It takes three iterations to find the fixed point.

Now you can check the result of the calculated reaching definitions.